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JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME	

CLASS		INDEX NUMBER					

BIOLOGY **9744/03**

Paper 3 Long Structured and Free-response Questions **11 September 2017**

Candidates answer on the Question Paper. **2 hours**

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper.
Circle the question number of the question attempted.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
4 / 5	
Total	

This document consists of **19** printed pages and **3** blank pages.

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Section A

Answer **all** the questions in this section.

1 The *lac* operon is an operon required for the uptake and metabolism of lactose in *Escherichia coli* and many other bacteria. Glucose is the preferred carbon source for most bacteria, as glucose requires fewer steps and less energy to break down than lactose. However, if lactose is the only sugar available, the *E. coli* uses it as an energy source and the *lac* operon allows for the effective digestion of lactose when glucose is not available.

To use lactose, the bacteria must express the *lac* operon genes, which encode key enzymes for lactose uptake and metabolism. Fig. 1.1 shows the results of an experiment carried out to determine the effects of adding lactose on the expression of some of the genes involved in the breakdown of lactose.

The initial gene expression was measured by determining the mRNA produced at time 0. This was taken as 10% (black bars). All the other values are relative to this initial value taken every 30 seconds over the next 2 minutes.

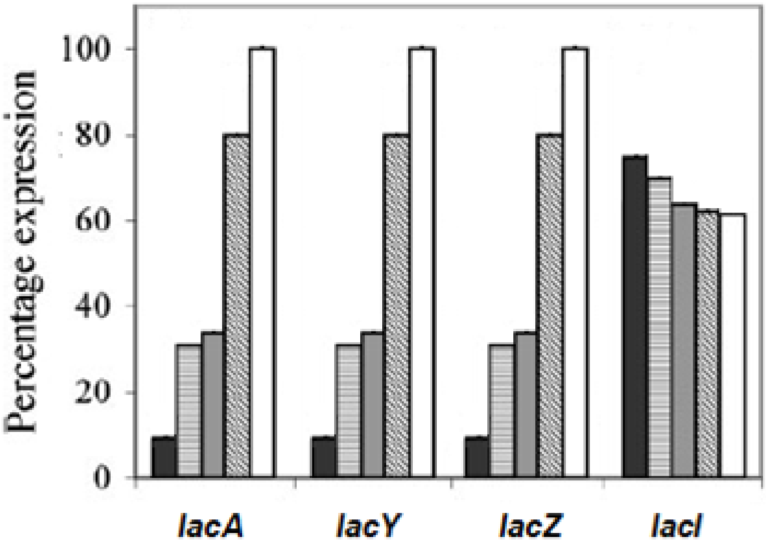

Fig. 1.1

(a) Suggest why operons are necessary in bacteria. [2]

(b) Using data from *lacA*, *lacY* and *lacZ* in Fig. 1.1 and your knowledge of how different types of operons are regulated, explain how lactose is able to control the expression of these genes. [4]

It has been suggested that not all genes involved in lactose hydrolysis are organised into one single operon.

(c) Use evidence from Fig. 1.1 to support the statement above. [3]

(d) A series of mutations was introduced into the *lac* operon, resulting in the inversion of the operator and the promoter regions.

Suggest the effect on the transcription of the *lac* genes when lactose is absent. [2]

Owing to *E. coli's* rapid growth rate, *E. coli* has been an expression host of choice in the biotechnology industry for large-scale production of anti-freeze proteins (AFPs). AFPs is a class of polypeptides that help to stop ice forming inside the Arctic and Antarctic fishes thus permitting their survival in sub-zero environments.

In the Arctic and Antarctic, environmental temperatures can reach low to freezing levels. These fishes indigenous to these habitats are presented with potential desiccation, which can lead to potentially detrimental challenges such as decreased enzymatic rates and freezing. Besides hindering cellular processes, sub-zero temperatures induce ice crystals formation, which can lead to cell death by rupturing cells either physically or through osmotic pressure changes.

Commercially, there appears to be countless applications for AFPs:

- as additives to frozen foods to lengthen the shelf life
- for incorporation with the genome of the raw foods to retard ice crystal growth
- to prevent damage to agricultural crops by increasing freeze tolerance of crop plants and extending the harvest season in cooler climates
- for introduction into ice cream and yogurt products to allow the production of very creamy, dense, reduced fat ice cream with fewer additives.

(e) Outline how the genome of *E. coli* and the genome of the fish are similar and how they are different. [4]

(f) Anti-freeze glycoprotein (AFGP) is one type of anti-freeze protein. Messenger RNA coding for AFGP is translated at a ribosome to produce a polypeptide. Describe how this polypeptide is then processed to make AFGP. [4]

Some fish produce another anti-freeze protein, called AFP II. The tissues of these fish were tested for the presence of AFP II and the mRNA coding for AFP II. The results are shown in Table 1.1.

Table 1.1

molecule	present in
AFP II protein	all tissues
AFP II mRNA	liver tissue only

(g) Explain the distribution of the AFP II protein and AFP II mRNA. [4]

(h) With reference to named examples, describe the roles performed by proteins involved in transport in fishes. [2]

Table 1.2 shows Earth's ice ages over the last 850 million years.

Table 1.2

Ice age	Time / millions of years ago
Quaternary	0 to 2.6
Karoo	260 to 360
Andean-Saharan	420 to 460
Cryogenian	630 to 850

Fig. 1.2 shows how the number of families of fishes has changed over time.

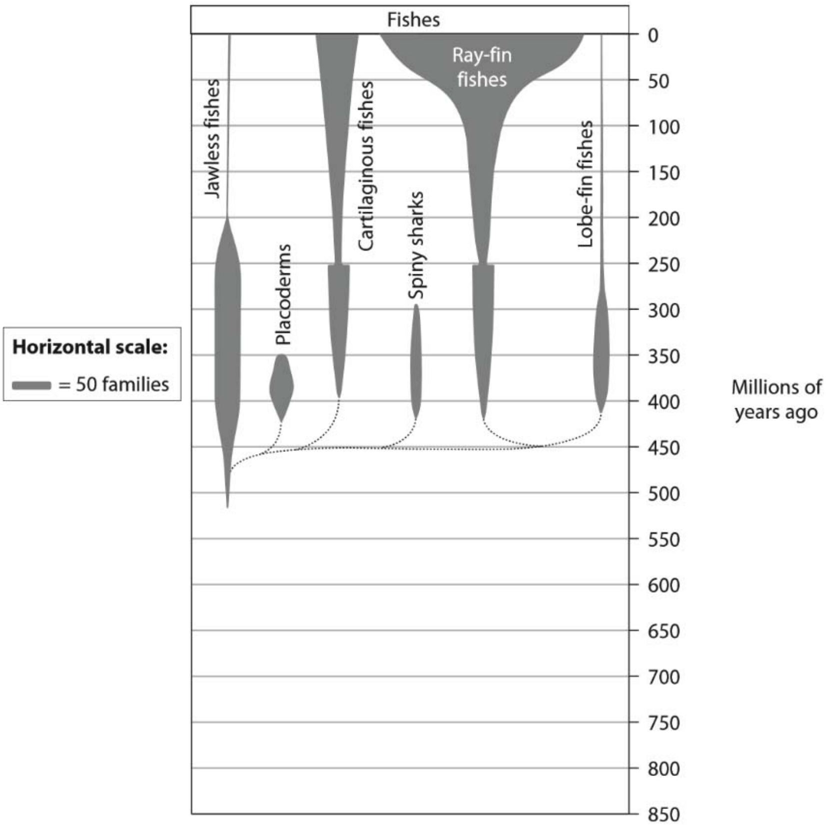

Fig. 1.2

(i) Many different types of AFPs are produced by ray-fin fishes. Analyse the data to explain when these ray-fin fishes are likely to have evolved the ability to produce AFPs. [2]

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[Total: 27]

2 Measles is a highly contagious, serious disease caused by *Morbillivirus*, a single-stranded enveloped RNA virus. Envelope glycoproteins mediate transmission of the virus into host cells in the human respiratory tract. Once inside the host cell, the viral RNA genome is transcribed into mRNA, which undergoes translation to manufacture viral proteins. These viral proteins function to form capsid proteins for new viruses which eventually leave the host cell.

(a) With reference to the information given, outline how viruses challenge the concept of what is considered living. [2]

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In the 1980s, measles caused an estimated 2.6 million deaths each year, and the disease remains one of the leading causes of death among young children globally.

The number of cases of measles is reported to the World Health Organisation (WHO) by countries throughout the world so that global data is collected.

Fig. 2.1 shows the global data collected between January 2008 and December 2012.

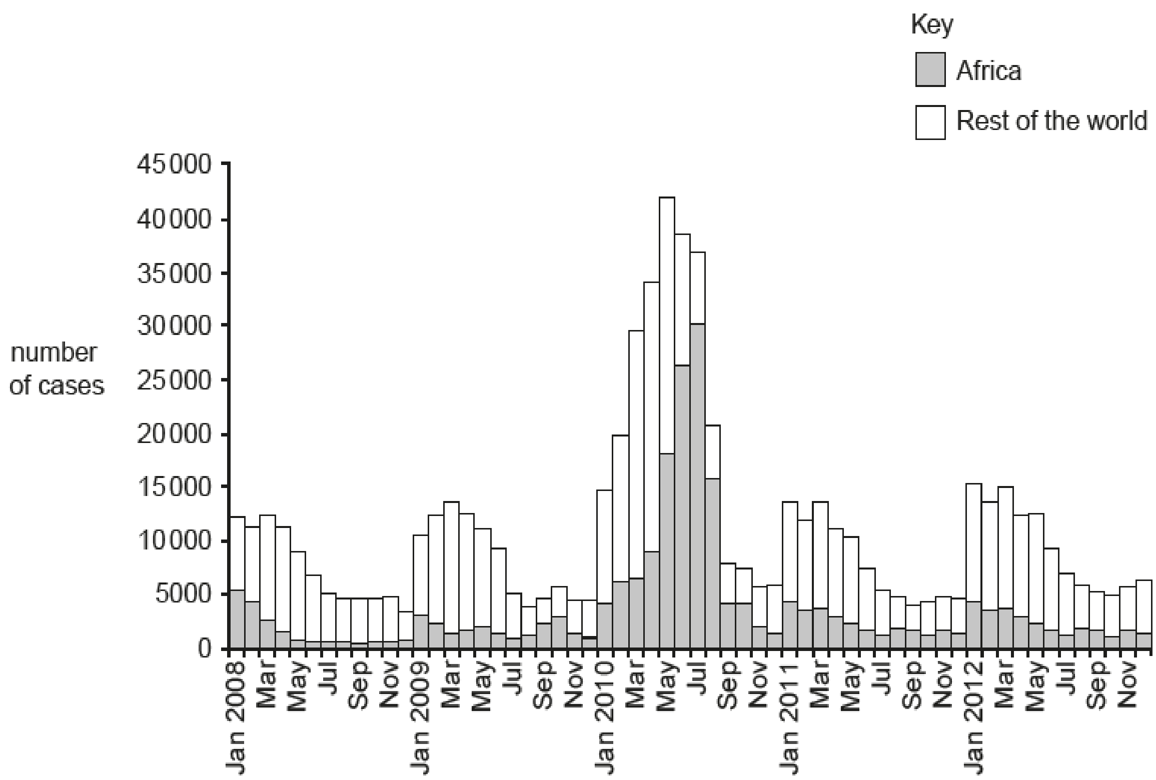

Fig. 2.1

(b) Use the data in Fig. 2.1 to describe the pattern shown in the number of cases of measles reported to the WHO between January 2008 and December 2012. [3]

(c) Routine measles vaccination for children, combined with mass immunisation campaigns in countries with high case and death rates, are key public health strategies to reduce global measles deaths. By 2016, about 85% of the world's children received one dose of the measles vaccine by their first birthday, and the global push to improve vaccine coverage resulted in a 79% reduction in deaths.

(i) State precisely the type of immunity gained by receiving a measles vaccine. [1]

(ii) Outline one benefit of vaccination. [1]

(iii) Outline one risk of vaccination. [1]

(d) Unlike measles for which an effective vaccine has been developed, it has been extremely difficult to design an effective vaccine against malaria. Malaria is a disease caused by the parasite *Plasmodium falciparum*. *P. falciparum* multiplies in liver cells of the host before emerging after 9-30 days wrapped in the liver cell surface membrane. They enter red blood cells, multiply and then cause rupture of the host cells, resulting in the release of more parasites every 36-48 hours, in a manner that has some similarity to that of viruses.

Use the information given and your own knowledge to suggest why it has been extremely difficult to design an effective vaccine against malaria. [2]

(e) Another infectious disease, Tuberculosis (TB), is one of the top ten causes of death worldwide. Name the bacterium that causes TB and describe how TB is transmitted. [3]

[Total: 13]

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3 Dengue fever is a disease spread by a particular species of mosquito, *Aedes aegypti.* The incidence of this disease and the numbers of this species of mosquito have increased dramatically in recent years, spreading beyond the tropics. This has been attributed to global warming.

(a) Explain how global warming has resulted in the spread of dengue beyond the tropics. [2]

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In an attempt to reduce the numbers of *A. aegypti*, male mosquitoes infected with the *Wolbachia* bacteria have been produced and released into the wild to mate with females. *Wolbachia* naturally occurs in up to 60% of all insect species, but not in *A. aegypti. Wolbachia* induces a conditional sterility that occurs within the mosquitoes due to cytoplasmic incompatibility, shown in Fig. 3.1, a concept first introduced in a paper published by Dr. Hannes Laven in 1967.

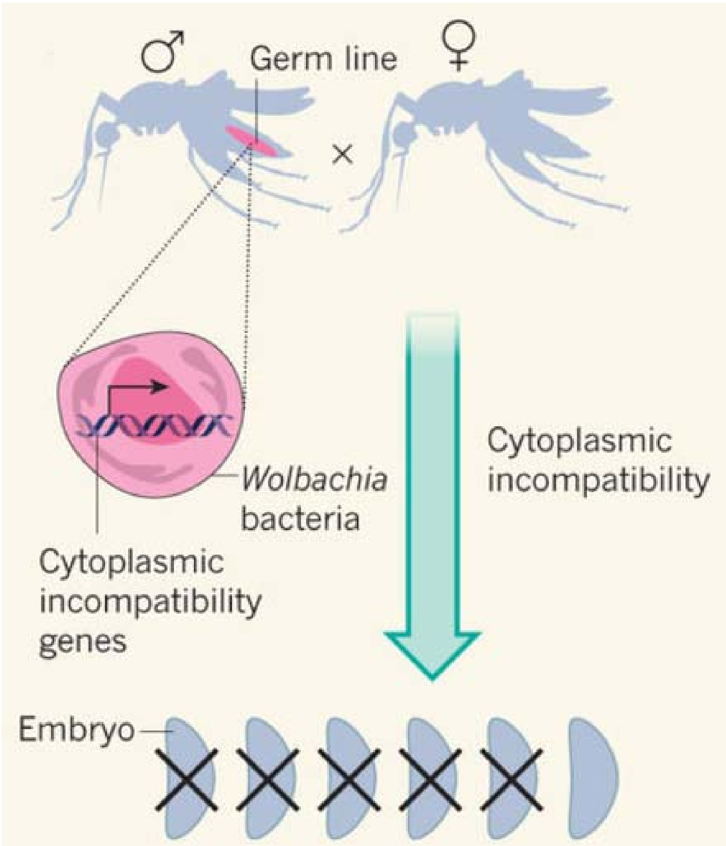

Fig. 3.1

(b) The cytoplasmic incompatibility genes are DNA in nature. DNA is a double helix consisting of two polynucleotide strands held together by phosphodiester bonds between the adjacent nucleotides. Each strand contains a sugar-phosphate backbone and hydrogen bonds are formed between the complementary strands via complementary base pairing.

Describe two other structural features of DNA. [2]

There are three possible matings between the male and female mosquitoes in the wild as shown in Table 3.1.

Table 3.1

	male	female	results
cross 1	infected	uninfected	lay eggs that are not viable and do not hatch
cross 2	infected	infected	infected offspring
cross 3	uninfected	infected	infected offspring

Embryonic development aborts when sperm from an infected male fertilises an uninfected egg and the paternal genome does not contribute to the development of the embryos that are not viable.

(c) Based on the information provided and Cross 1, suggest how *Wolbachia* induces sterility. [1]

In 1970, Erich Jost repeated Laven's earlier work.

(d) Explain why repeating the work of others is an important part of science research. [2]

In 2016, hundred thousands of male mosquitos with *Wolbachia* were released at 3 selected sites in Singapore: Braddell Heights, Nee Soon East, and Tampines West.

(e) State why releasing such large numbers of male mosquitoes did not immediately increase the risk of transmission of dengue fever in these estates. [1]

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Another method employed in Australia involves the release of both male and female mosquitoes with *Wolbachia* into the wild. An advantage of this method is that there is no need for further releases of mosquitoes with *Wolbachia.*

(f) With reference to Table 3.1, explain why there is no need for further releases with this method. [2]

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[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)**, **(b)** etc., as indicated in the question.

4 **(a)** Describe the reproductive cycle of the influenza virus and explain how new strains of the virus may arise as a result of mutation. [13]

(b) Describe the process of transduction and its advantages to prokaryotes. [12]

[Total: 25]

5 **(a)** Describe the roles of the proteins involved in the process of DNA replication and compare the advantages of PCR with the advantages of DNA replication. [13]

(b) Outline the structure of G-protein linked receptor and describe the action of glucagon on liver cells in the regulation of blood glucose concentration. [12]

[Total: 25]

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